

● HARD

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

02

Q1

GRID-IN ADVANCED MATH

The function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. What is the product of $g15a$ and $g7a$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$fx = 1.84^{\frac{x}{4}}$$

The function f is defined by the given equation. The equation can be rewritten as $fx = 1 + \frac{p}{100}^x$, where p is a constant. Which of the following is closest to the value of p ?

- A. 16
- B. 21
- C. 46
- D. 96

$$fx = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

- A. $fx = -6x^2 + 42x + 18x - 126$
- B. $fx = -6x(x - 7) + 18(x - 7)$
- C. $fx = -6x - 5^2 + 24$
- D. $fx = -6x - 7x - 3$

Q4

ADVANCED MATH

For the function q , the value of qx decreases by 45% for every increase in the value of x by 1. If $q0 = 14$, which equation defines q ?

A. $qx = 0.5514^x$

B. $qx = 1.4514^x$

C. $qx = 140.55^x$

D. $qx = 141.45^x$

Q5

ADVANCED MATH

$$fx = x^2 - 14x + 19$$

The function f is defined by the given equation. For what value of x does fx reach its minimum?

A. -266

B. -19

C. $-\frac{33}{2}$

D. $-\frac{5}{2}$

Let the function p be defined as $p(x) = \frac{(x-c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

- A. 10.00
- B. 10.25
- C. 10.75
- D. 11.00

$$f(t) = 8,000(0.65)^t$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of 1998, and $0 \leq t \leq 5$. If $y = f(t)$ is graphed in the ty -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

- A. The minimum estimated number of coupons the company sent to their customers during the 5 years was 1,428.
- B. The minimum estimated number of coupons the company sent to their customers during the 5 years was 8,000.
- C. The estimated number of coupons the company sent to their customers at the end of 1998 was 1,428.
- D. The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

$$p(t) = 90,000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \frac{90,000}{12}(1.06)^m$

B. $r(m) = 90,000 \frac{1.06^m}{12}$

C. $r(m) = 90,000 \frac{1.06^{\frac{m}{12}}}{12}$

D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8 ?

- A. Day 5
- B. Day 9
- C. Day 11
- D. Day 12

Which of the following functions has(have) a minimum value at -3?

I. $f(x) = -6(3)^x - 3$

II. $g(x) = -3(6)^x$

A. I only

B. II only

C. I and II

D. Neither I nor II